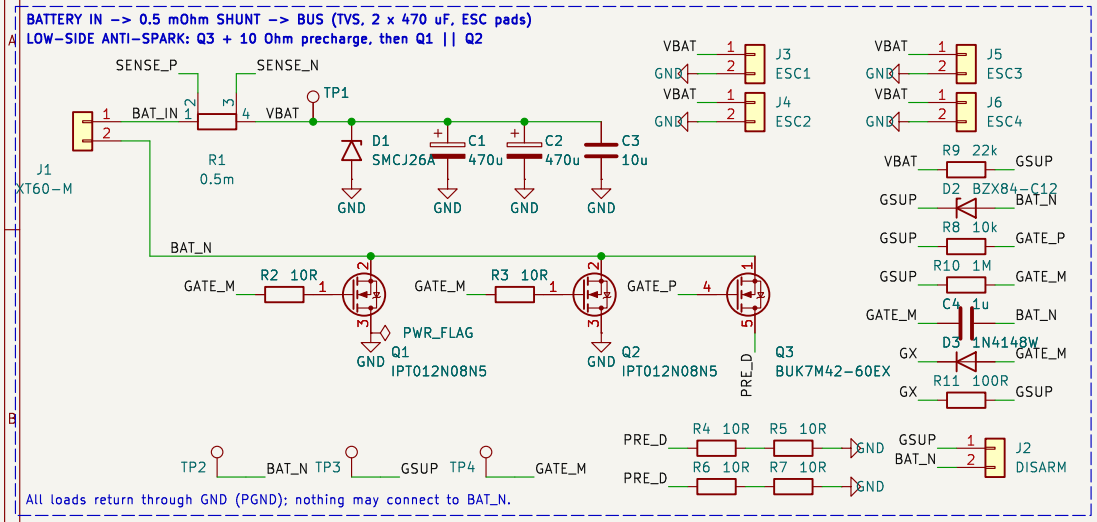
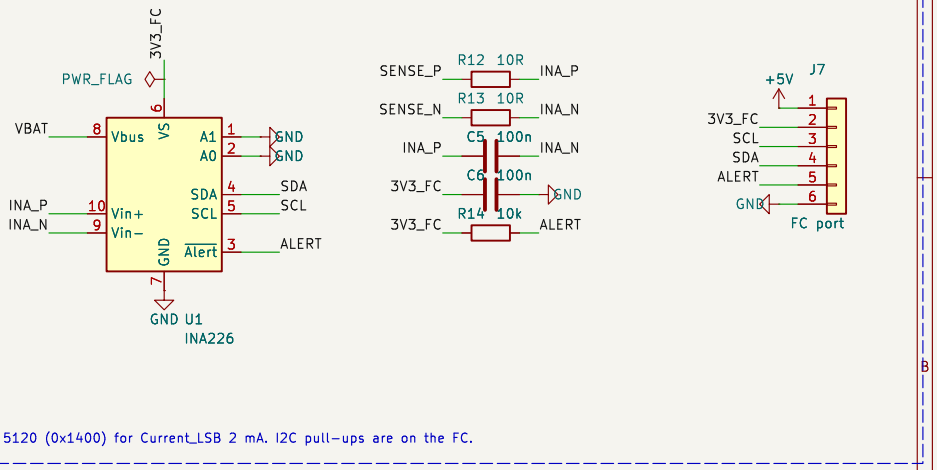
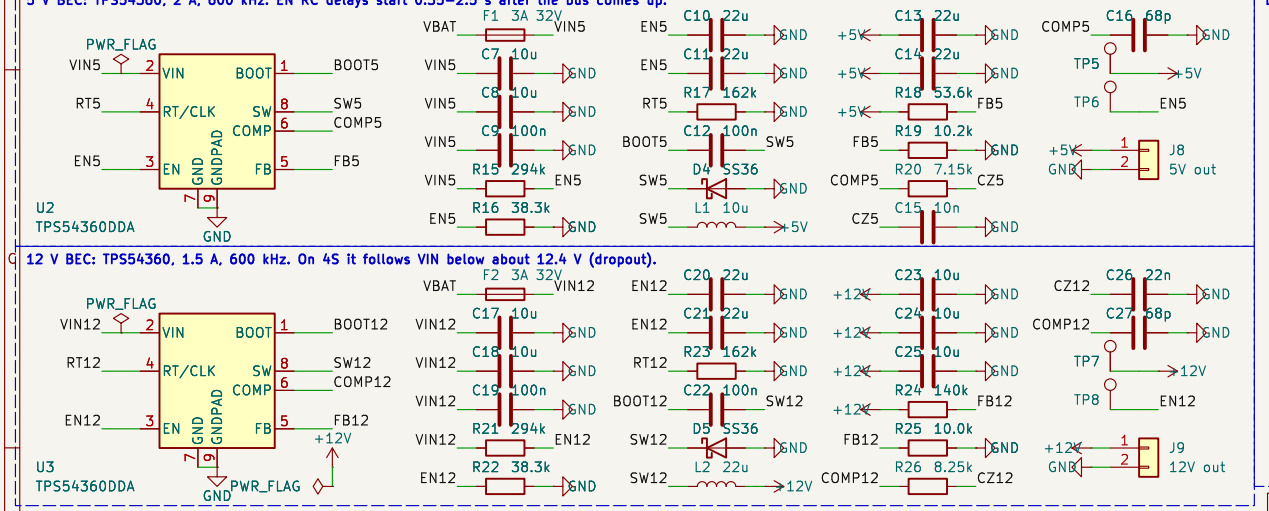
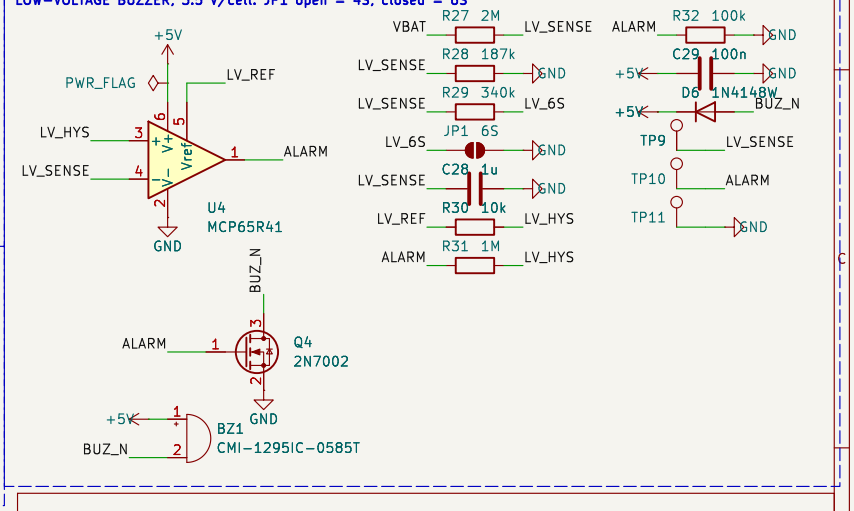
WARNING: Reference design. Test first with a current-limited supply and protection. Not for unattended use.

Design only, not fabricated, not bench-tested. LiPo/Li-ion drone packs: this board has no BMS function; use packs with their own protection. Never charge or store a pack through this board.









KEY VALUES (README has the derivations and sources)

30 A continuous / 60 A burst (XT60). Q1 || Q2 0.6 mOhm max at VGS 10 V. Shunt 0.5 mOhm, 1 W.

Precharge 10 Ohm x 2200 uF (assumed) = 22 ms; 2.5 A peak, 0.70 J in R4-R7 at 25.2 V (pulse rating unverified).

Main FETs: R10 1M x C4 1u, VGS(th) after 174-465 ms, 10 V after about 1.9 s. BECs start after the main FETs conduct.

J2 DISARM plug shorts GSUP: Q1-Q3 off. Disarm only with motors stopped. Disarmed draw about 1.2 mA at 25 V.

SMCJ26A clamps at 42.1 V at full Ipp: above INA226 40 V abs max (TPS54360 65 V is fine).

LV alarm ON below 14.0 V (4S) / 21.1 V (6S), OFF above 14.6 / 21.9 V; worst case +-0.7 / +-1.0 V.

XT60 pin 1 = BAT+ is assumed: check footprint pad 1 against the housing before layout.

Generated from stock KiCad 9 library symbols by scripts/gen_schematic.py

Hardware licence: CERN-OHL-P-2.0

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Abdulazeez Bello

Sheet: /

File: drone-pdb.kicad_sch

Title: Drone power board: 4S/6S, anti-spark, INA226, BECs

Size: A4	Date: 2026-10-18	**Rev: A**
KiCad E.D.A. 9.0.3		Id: 1/1